

Module 3 DataBase Integration

Hands on -7: Alembic migrations

Task 1:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic init migrations
Creating directory C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations ... done
Creating directory C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions ... done
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\alembic.ini ... done
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\env.py ... done
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\README ... done
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\script.py.mako ... done
Please edit configuration/connection/logging settings in C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\alembic.ini before proceeding.
```

Initial schema:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic revision --autogenerate -m "initial schema"
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.schemas
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.tables
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.types
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.constraints
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.defaults
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.comments
INFO [alembic.autogenerate.compare.tables] Detected removed table 'department_transfer_log'
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\550b0656bfbf_initial_schema.py ... done
```

Upgrade head:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic upgrade head
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Running upgrade -> 550b0656bfbf, initial schema
```

Task 2:

Add new column to employees

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic revision --autogenerate -m "add is_active to employees"
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.schemas
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.tables
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.types
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.constraints
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.defaults
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.comments
INFO [alembic.autogenerate.compare.tables] Detected added column 'employees.is_active'
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\e0171c4aeeed_add_is_active_to_employees.py ... done
```

Upgrade head:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic upgrade head
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Running upgrade 550b0656bfbf -> e0171c4aeeed, add is_active to employees
```

New table module_schedule:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic revision --autogenerate -m "add ModuleSchedule table"
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\.\models.py:71: SAWarning: Can't validate argument 'auto_increment'; can't locate any SQLAlchemy dialect name
d 'auto'
      schedule_id=Column(Integer,primary_key=True,auto_increment=True)
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.schemas
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.tables
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.types
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.constraints
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.defaults
INFO [alembic.runtime.plugins] setting up autogenerate plugin alembic.autogenerate.comments
Generating C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\882dadd59d88_add_moduleschedule_table.py ... done
```

Upgrade head:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic upgrade head
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\.\models.py:71: SAWarning: Can't validate argument 'auto_increment'; can't locate any SQLAlchemy dialect name
d 'auto'
      schedule_id=Column(Integer,primary_key=True,auto_increment=True)
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Running upgrade e0171c4aeecd -> 882dadd59d88, add ModuleSchedule table
```

History:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic history --verbose
Rev: 882dadd59d88 (head)
Parent: e0171c4aeecd
Path: C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\882dadd59d88_add_moduleschedule_table.py

    add ModuleSchedule table

    Revision ID: 882dadd59d88
    Revises: e0171c4aeecd
    Create Date: 2026-06-21 20:06:53.485937

Rev: e0171c4aeecd
Parent: 550b0656bfbf
Path: C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\e0171c4aeecd_add_is_active_to_employees.py

    add is_active to employees

    Revision ID: e0171c4aeecd
    Revises: 550b0656bfbf
    Create Date: 2026-06-21 19:59:56.817868

Rev: 550b0656bfbf
Parent: <base>
Path: C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\migrations\versions\550b0656bfbf_initial_schema.py

    initial schema

    Revision ID: 550b0656bfbf
    Revises:
    Create Date: 2026-06-21 19:44:54.144592
```

Task 3:

Current head:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic current
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6\.\models.py:71: SAWarning: Can't validate argument 'auto_increment'; can't locate any SQLAlchemy dialect name
d 'auto'
      schedule_id=Column(Integer,primary_key=True,auto_increment=True)
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
882dadd59d88 (head)
```

Rollback:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic downgrade -1
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>.\models.py:71: SAWarning: Can't validate argument 'auto_increment'; can't locate any SQLAlchemy dialect name
d 'auto'
  schedule_id=Column(Integer,primary_key=True,auto_increment=True)
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Running downgrade 882dadd59d88 -> e0171c4aeecd, add ModuleSchedule table
```

Re apply migrations:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>alembic upgrade head
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module3_DatabaseIntegration\VigneshJaganathan\HandsOn6>.\models.py:71: SAWarning: Can't validate argument 'auto_increment'; can't locate any SQLAlchemy dialect name
d 'auto'
  schedule_id=Column(Integer,primary_key=True,auto_increment=True)
INFO [alembic.runtime.migration] Context impl MySQLImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Running upgrade e0171c4aeecd -> 882dadd59d88, add ModuleSchedule table
```

Same head has as before rollback.